Machine Learning Internship

Project Name: Multi-Label Emotion Classification

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1. Executive Summary

This project presents a deep learning-based solution for multi-label emotion classification in short English text. The objective is to automatically identify multiple emotions—anger, fear, joy, sadness, and surprise—from a given text input. To achieve this, a transformer-based architecture using roberta-base is implemented and fine-tuned on the dataset. The model leverages contextual embeddings to capture subtle emotional cues in text.

The system incorporates several advanced techniques, including:

Class imbalance handling using weighted binary cross-entropy loss
Early stopping to prevent overfitting
Learning rate scheduling with warm-up
Threshold optimization for improved multi-label prediction

The final model outputs probabilistic predictions for each emotion, which are converted into binary labels using optimized thresholds. This approach ensures improved macro F1-score performance, making the model suitable for real-world emotion detection tasks.

2. Introduction

2.1 Problem Statement

In natural language processing, understanding human emotions from text is a complex challenge due to:

- The subjective nature of emotions
- Overlap of multiple emotions in a single sentence
- Short text length, limiting contextual information

This project addresses the problem of:

Accurately classifying short English text into multiple emotion categories (anger, fear, joy, sadness, surprise), where each text can express more than one emotion simultaneously.
The task is formulated as a multi-label classification problem, where each label is independently predicted.

2.2 Project Objectives

The primary goal of this project is to develop a robust deep learning model capable of identifying multiple emotions from short text inputs. To achieve this, a pre-trained transformer model, roberta-base, is utilized to effectively capture contextual and semantic meaning. The project addresses the multi-label nature of the problem by implementing sigmoid activation and BCEWithLogitsLoss, allowing the model to predict multiple emotions simultaneously. To handle class imbalance, class-weighted loss (pos_weight) is applied, ensuring better performance on underrepresented emotion categories. Model performance is further optimized using techniques such as learning rate scheduling, gradient clipping, and early stopping. Additionally, prediction accuracy is improved through threshold optimization for each label to maximize the macro F1-score. Finally, an end-to-end pipeline is developed, including training, validation, and inference stages, to ensure a complete and practical solution.

2.3 Report Structure

This project begins with an introduction to emotion detection in natural language processing, highlighting the importance of multi-label classification and its real-world applications such as chatbots, sentiment analysis, and mental health monitoring. The problem is defined using a dataset consisting of short text entries labeled with multiple emotions, where each label represents a specific emotion category, making it a multi-label classification challenge. The methodology includes data preprocessing steps like text cleaning and train-validation splitting, followed by a model architecture based on the pre-trained roberta-base, incorporating a dropout layer and a fully connected classification head. The training strategy uses BCEWithLogitsLoss with class weights, AdamW optimizer, learning rate scheduling with warm-up, and early stopping to improve performance. Evaluation is conducted using macro F1-score along with threshold-based methods, and further enhanced through per-label threshold optimization using grid search. The implementation is carried out using PyTorch and Hugging Face Transformers, with well-structured dataset handling and training pipelines. Results and analysis focus on validation F1-score trends and the impact of threshold tuning. Finally, the project concludes by summarizing outcomes, strengths, and limitations, and suggests future improvements such as using advanced models like DeBERTa, applying data augmentation, ensemble techniques, and deploying the model in real-time applications.

3. Dataset & Preprocessing

3.1 Dataset Description

The training set contains 6827 labeled texts, and the test set contains 1707 unlabeled texts. Each row consists of an id, the text, five binary columns for the emotions, and a derived list of active emotions. The label distribution is imbalanced: fear is the most frequent emotion, followed by sadness, surprise, joy, and anger, which is the least represented. Many samples contain multiple emotions simultaneously, and a non-trivial number have no active emotion label at all, which adds difficulty to the task.

3.2 Data Preprocessing

The preprocessing step is designed to preserve semantic meaning while ensuring clean and consistent input for the model. Initially, text cleaning is performed using the `clean_text()` function, which removes leading and trailing spaces, normalizes multiple spaces into a single space, and converts all inputs into string format. Unlike traditional NLP pipelines, the text is not lowercased, as the **roberta-base** is case-sensitive and benefits from preserving capitalization for better contextual understanding. Finally, the dataset is split into training and validation sets, with 90% used for training and 10% for validation, and random shuffling is applied to ensure better generalization of the model.

4. Tokenization Strategy

Tokenization is the process of converting raw text into numerical representations that can be processed by the transformer model. In this project, a **subword tokenization** approach is used, implemented through the Hugging Face tokenizer using `AutoTokenizer.from_pretrained(MODEL_NAME)`. The primary tokenizer selected is the one associated with the **roberta-base**, which is specifically designed to work with the RoBERTa architecture. This tokenizer uses **Byte-Pair Encoding (BPE)** and a pre-trained vocabulary aligned with the RoBERTa model, ensuring that the tokenized inputs are compatible with the model's learned embeddings and enabling effective capture of contextual and semantic information from the text.

4.1 Primary Tokenizer Selection

The primary tokenizer selected for this project is the tokenizer associated with the pre-trained roberta-base. This tokenizer is specifically designed to align with the RoBERTa model architecture, ensuring that the input text is converted into token IDs that match the model's pre-trained embeddings. It is based on Byte-Pair Encoding (BPE), a subword tokenization technique that efficiently handles unknown and rare words by breaking them into smaller meaningful units. This helps preserve important contextual and semantic information, especially in short text where emotional cues are subtle. By using the RoBERTa tokenizer, the project ensures consistency between preprocessing and model training, leading to better performance and more accurate emotion classification.

4.2 Justification for Tokenization Choice

The tokenizer selected for this project is the one associated with roberta-base, which is based on Byte-Pair Encoding (BPE). This choice is primarily justified by its compatibility with the pre-trained RoBERTa model, ensuring that the token representations align correctly with the model's learned embeddings. Unlike traditional word-level tokenization, BPE effectively handles rare and unseen words by decomposing them into meaningful subword units, thereby reducing information loss. This is particularly important for emotion detection tasks where subtle variations in words can significantly impact meaning. Additionally, RoBERTa's tokenizer preserves case sensitivity, which helps capture emotional emphasis (e.g., "HAPPY" vs "happy"). It is also robust to informal language, typos, and short text patterns commonly found in real-world datasets. Overall, this tokenizer enhances the model's ability to capture contextual and semantic nuances, making it highly suitable for multi-label emotion classification.

4.3 Implementation Details

The tokenization process is implemented using the Hugging Face AutoTokenizer, initialized with the RoBERTa pre-trained model. Each input text is processed through the tokenizer with truncation and padding enabled, ensuring that all sequences are converted into a fixed length of 256 tokens. This standardization allows efficient batch processing during training and inference. The tokenizer generates two key outputs: input_ids, which represent the numerical encoding of tokens, and attention_mask, which distinguishes between actual tokens and padded elements. These outputs are returned as PyTorch tensors and integrated into a custom dataset class, where they are formatted for model consumption. During training, corresponding label tensors are also attached to each sample for multi-label learning. The processed data is then fed into PyTorch DataLoaders with appropriate batch sizes for training, validation, and testing. This implementation ensures seamless integration between preprocessing, model input requirements, and efficient GPU-based computation.

5. Modeling & Experimentation

5.1 RoBERTa-Based Model (Fine-Tuned Transformer 2)

5.1.1 Architecture Overview

The third model uses roberta-base(Figure 1) as the encoder, with a structure parallel to the BERT model:

- RobertaTokenizer converts text into BPE subword IDs and attention masks, again with max_length=200.
- RobertaModel encodes the sequence and returns a pooled representation.
- A dropout layer (0.3) is applied to the pooled output.
- A linear classifier projects the 768-dimensional pooled vector to five logits.
- A sigmoid at inference time yields multi-label emotion probabilities.

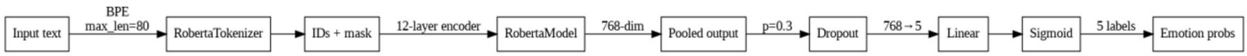

Figure 1: Roberta Architecture

5.1.2 Training Strategy

The training loop mirrors the BERT setup:

- Loss: BCEWithLogitsLoss.
- Optimizer: AdamW with learning rate 2e-5.
- Batch size: 16, maximum of 5 epochs with early stopping (patience 3) on validation macro-F1.
- 10% of the training data is held out for validation; threshold 0.5 is used for binarization.
- Metrics and loss are logged to Weights & Biases each epoch.

6. Performance & Comparative Analysis

6.1 Evaluation Metrics

The primary metric is macro-averaged F1 score, matching the Kaggle competition's scoring function. This metric separately computes F1 for each emotion and averages them, ensuring that performance on minority labels like anger is reflected fairly. Accuracy is also reported as a secondary metric but is less informative due to label imbalance and the multi-label setup.

6.2 Training Dynamics and Model Behaviour

The model is fine-tuned using the pre-trained roberta-base with the AdamW optimizer and a low learning rate for stable updates. A linear learning rate scheduler with warm-up ensures smooth convergence during training. Gradient clipping is applied to prevent instability, while class imbalance is handled using weighted BCEWithLogitsLoss. Early stopping based on validation F1-score helps avoid overfitting and preserves the best model. The model uses sigmoid activation to support multi-label predictions, allowing multiple emotions per text. Finally, threshold optimization improves prediction accuracy by selecting optimal decision boundaries for each emotion class.

7. Conclusion & Future Work

This project successfully developed a multi-label emotion classification system using the pre-trained roberta-base, achieving effective detection of multiple emotions in short text. The use of weighted loss, early stopping, and threshold optimization improved model performance and generalization. The model demonstrates strong capability in capturing nuanced emotional expressions and handling class imbalance. However, performance can still be enhanced for rare emotion categories.

For future work, advanced models such as DeBERTa can be explored to improve accuracy. Incorporating data augmentation techniques and larger datasets may further boost performance. Ensemble methods and hyperparameter tuning can also be applied for better results. Additionally, deploying the model as a real-time application using Streamlit or Hugging Face Spaces can make it more practical and user-friendly.

7.1 Key Learnings

This project provided strong practical exposure to transformer-based NLP using roberta-base. It helped in understanding multi-label classification, especially the use of sigmoid activation and BCEWithLogitsLoss instead of softmax. The importance of handling class imbalance using weighted loss and improving performance through threshold optimization was also learned. Additionally, experience was gained in building end-to-end pipelines including preprocessing, training, evaluation, and inference.

7.2 Challenges Faced

One major challenge was handling class imbalance, where some emotions were underrepresented, affecting model performance. Another difficulty was tuning hyperparameters such as learning rate, batch size, and epochs to achieve stable convergence. Optimizing thresholds for each label to improve macro F1-score required careful experimentation. Managing computational resources and training time, especially with transformer models, was also challenging.

7.3 Future Work

Future improvements include experimenting with advanced models like DeBERTa for better accuracy and exploring ensemble techniques. Data augmentation and collecting more balanced datasets can further enhance performance. Fine-tuning hyperparameters and using cross-validation may improve robustness. Finally, deploying the model using tools like Streamlit or Hugging Face Spaces can make the solution more practical and accessible.

8. References

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